How do annual members and casual riders use Cyclistic bikes differently?

Cyclistic case study · Share phase

October 2025 – September 2026 · 6.0 million trips · Divvy public data

Casual trips concentrate in summer, on weekends and at a few stations

40.8%

casual share of trips in July; 17.9% in January

52.9%

of casual trips happen Friday–Sunday (members: 38.0%)

+2.5

minutes longer median trip for casual riders (11.1 vs 8.7 min)

3.4×

round-trip share of casual trips vs. member trips

Shares are shares of trips, not of users. Clean data: 6,007,779 trips (10/2025 – 09/2026).

Task: find the differences that could convert casual riders to members

Business task

Help the marketing team design strategies to convert casual riders into annual members by showing how members and casual riders use the bikes differently.

Short answer

Casual trips concentrate in summer, on weekends and at a few stations; member trips peak on weekday mornings and evenings.



Data

Divvy public trip data (Cyclistic is a fictional company)



Period

October 2025 – September 2026, 12 months



Unit

Trips. No user IDs, so shares are shares of trips

From 6.17 million rows to 6.01 million rows of clean data

6,171,785
raw data (12 monthly tables)

→

−71
out-of-period rows

→

−35
duplicate records

→

−163,900
duration filter (outside 60 s – 24 h)

→

6,007,779
analysis table

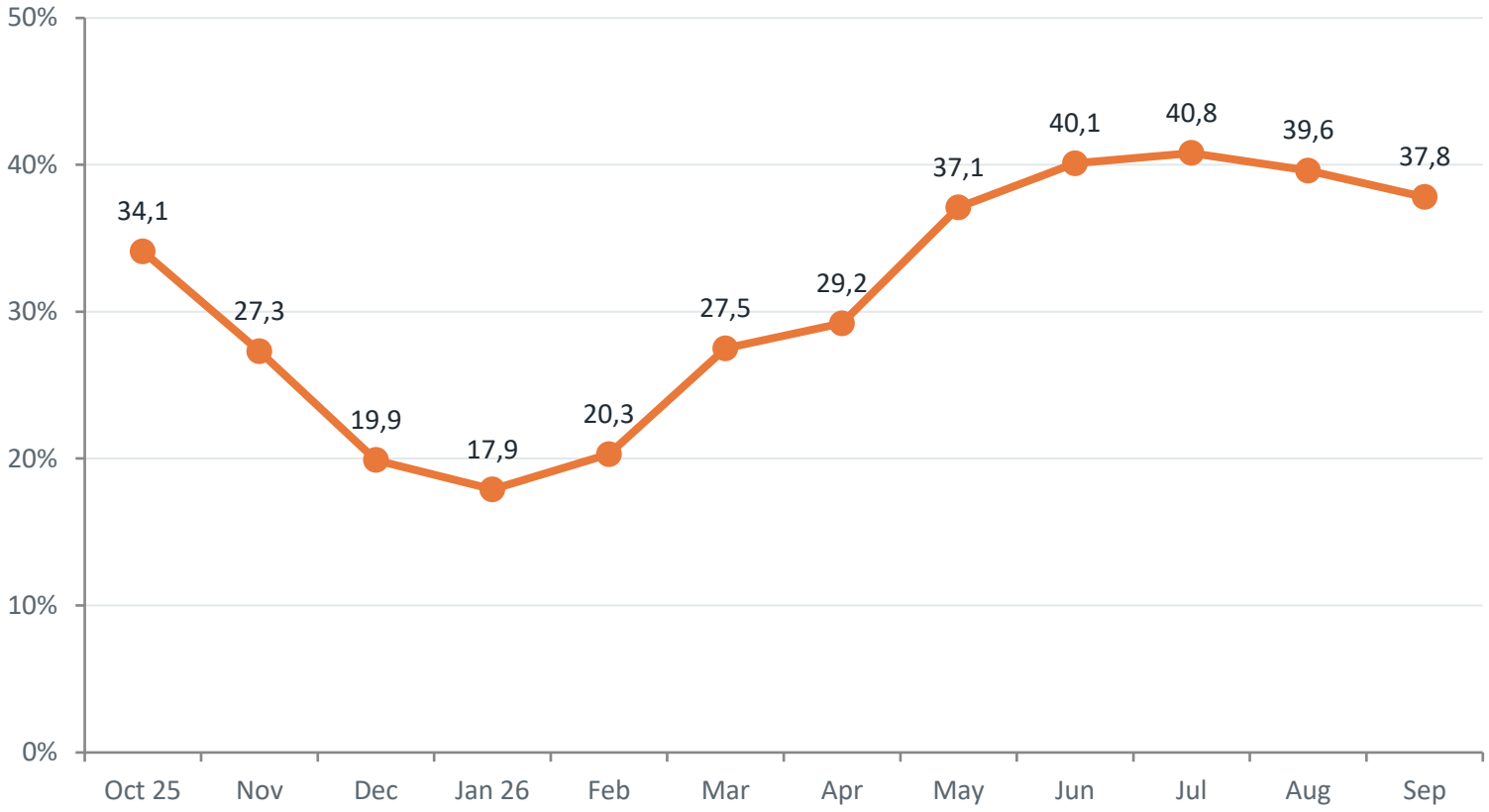
What we removed

- Out-of-period rows
- Exact duplicate records
- Trips shorter than 60 seconds or longer than 24 hours

What we kept and must watch

- Missing station info (about 20%, all electric bikes): excluded only from the station analysis
- The duration filter removed about 3.6% of casual trips and 2.1% of member trips

Casual share of trips rises to 40% in summer and falls to 18% in winter



17.9% → 40.8%

Casual share, January to July

14.5× · 4.6×

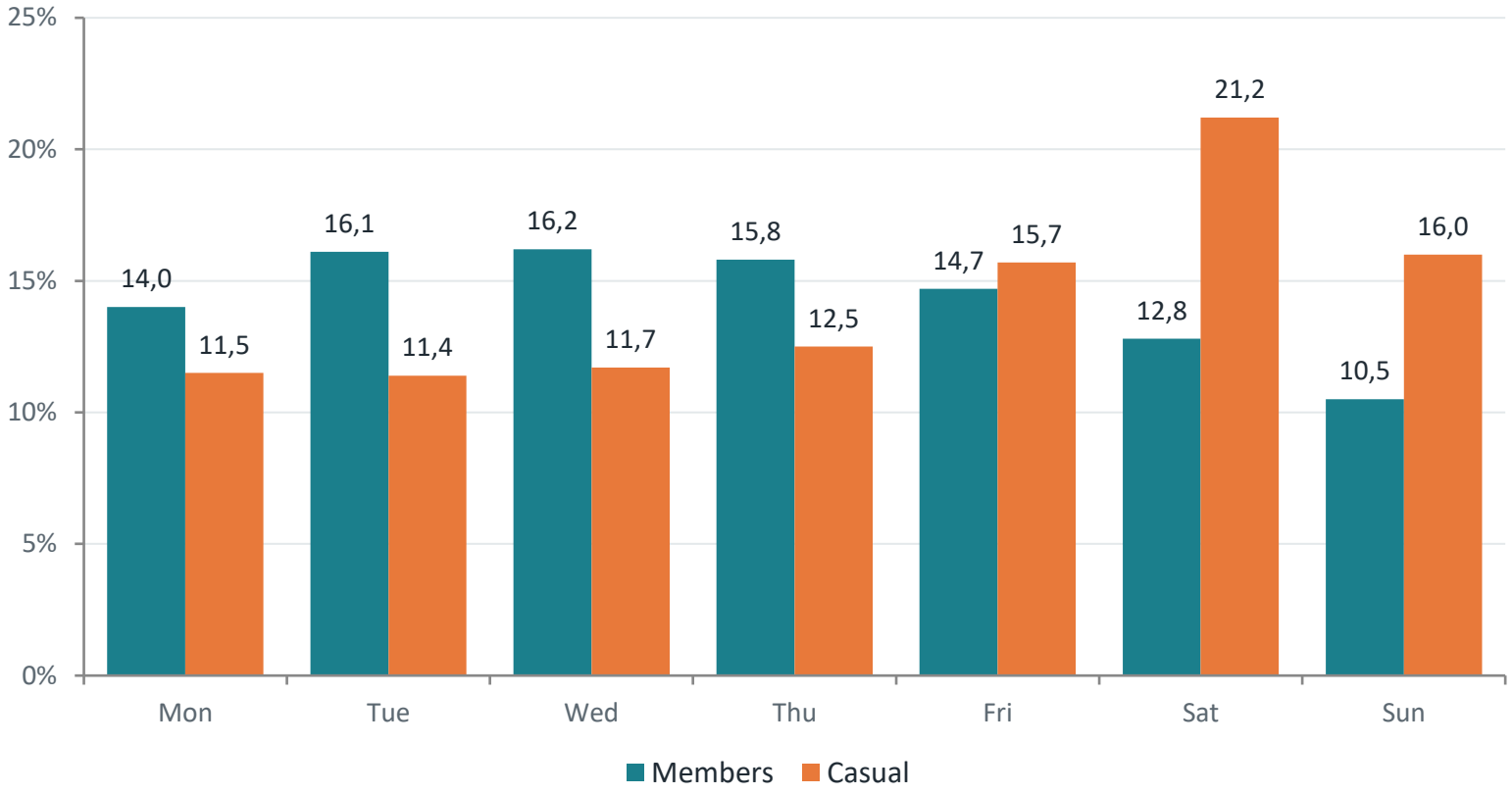
Growth in casual and member trips, January to July (approx.)

39.6% · 30.5%

Casual share, May–Aug and the other 8 months

The pre-set threshold (15 points between May–Aug and the other 8 months) was not met: the direction held, the difference was smaller than expected (about 9 points).

53% of casual trips fall on Friday–Sunday



52.9%

of casual trips happen Friday–Sunday

38.0%

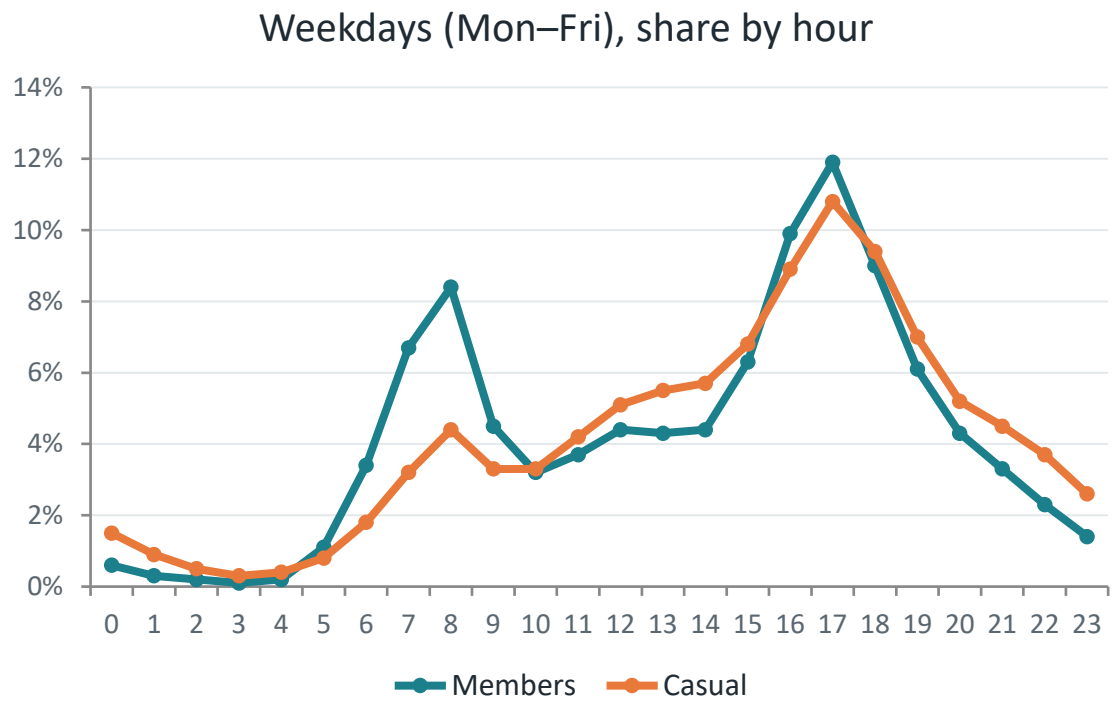
of member trips happen Friday–Sunday

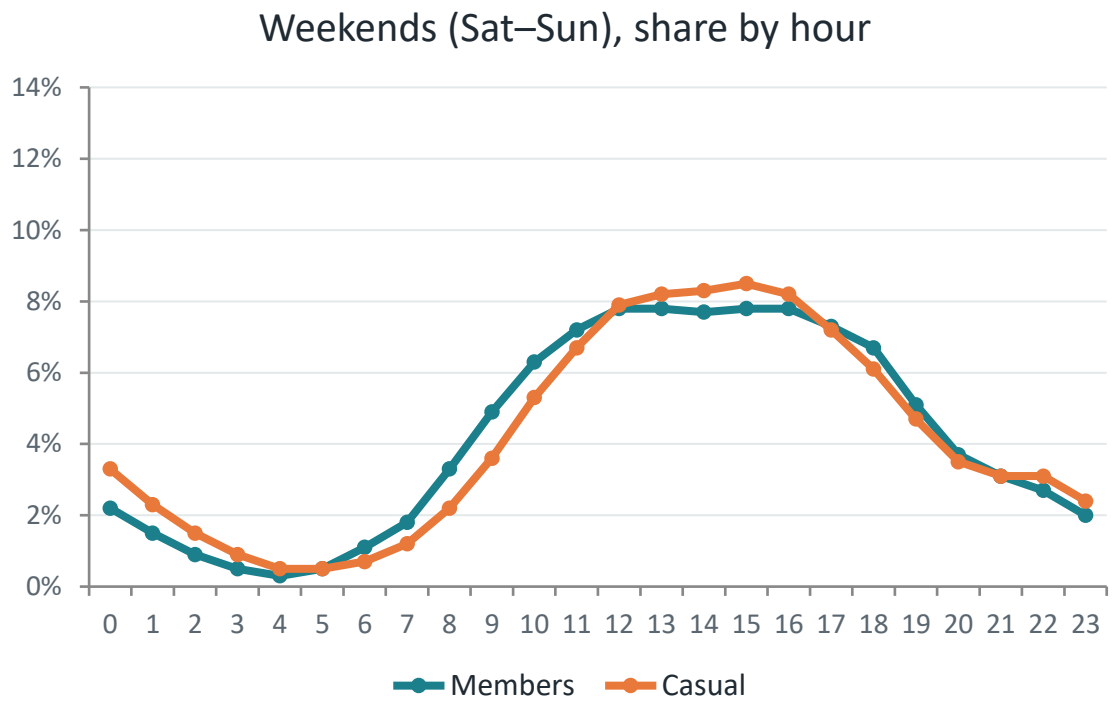
+13.9 pts

almost all of the gap is Saturday–Sunday; Friday is only +1.0

Percent within each group's trips. This comparison was made after seeing the table (exploratory observation).

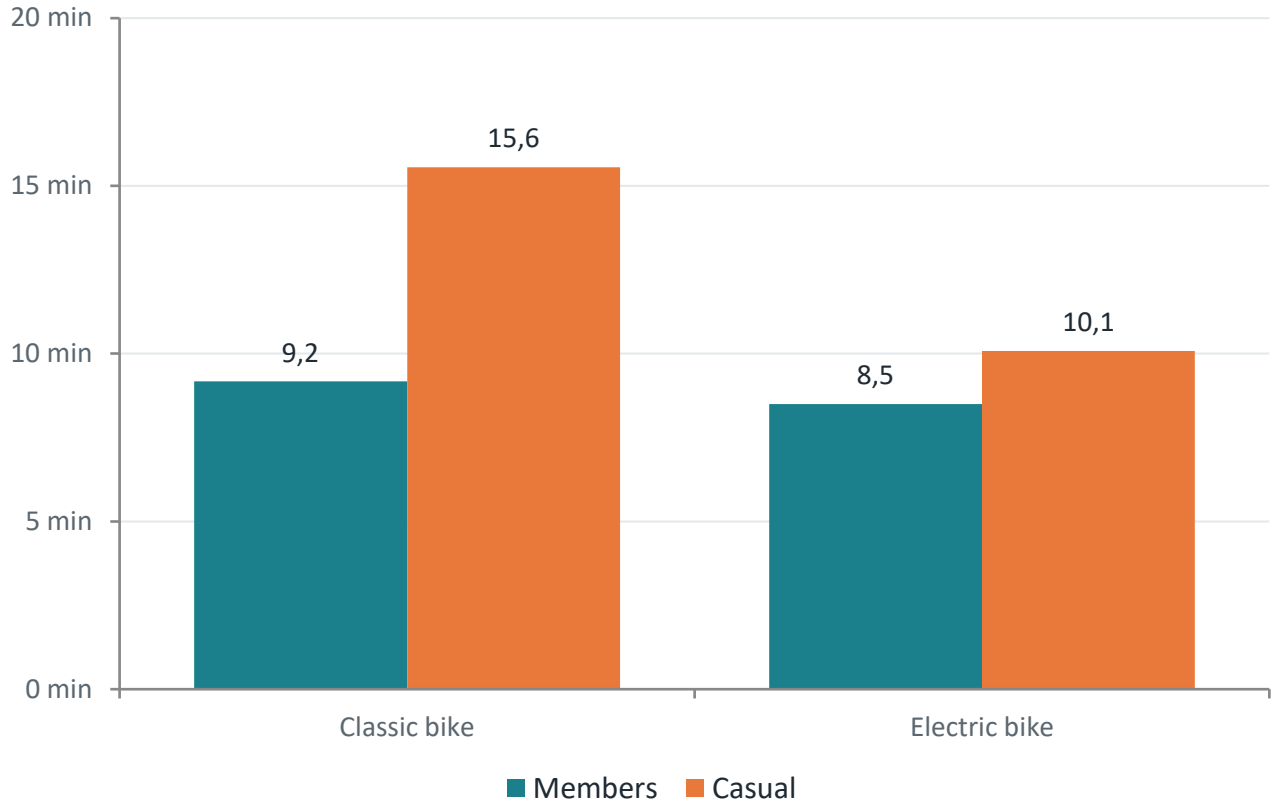
Members peak on weekday mornings and evenings; weekends show no peaks





Morning peak (08:00): members 8.4%, casual 4.4% (below casual's midday level). **Evening peak (17:00):** members 11.9%, casual 10.8%. *Timing does not show the purpose of a trip.*

Casual trips are longer on both bike types, most of all on classic bikes



Median trip duration (minutes). The median shows the typical trip; the mean is pulled up by long trips.

Mean duration, casual vs members

18.2 vs 12.2 min

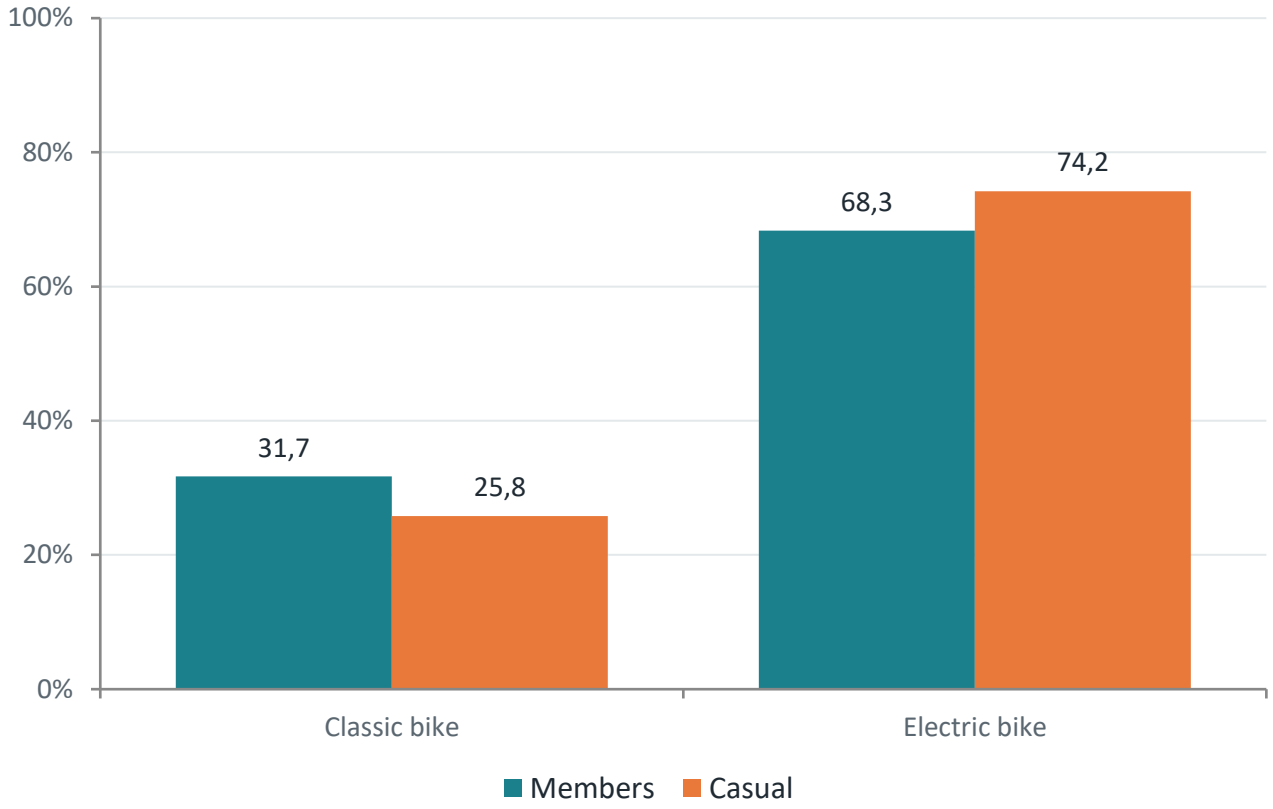
Median duration, casual vs members

11.1 vs 8.7 min

Trips over 60 min, casual vs members

3.9% vs 0.7%

74% of casual trips are made on electric bikes



Percent within each group’s trips.

74.2% · 68.3%

electric bike share: casual and members

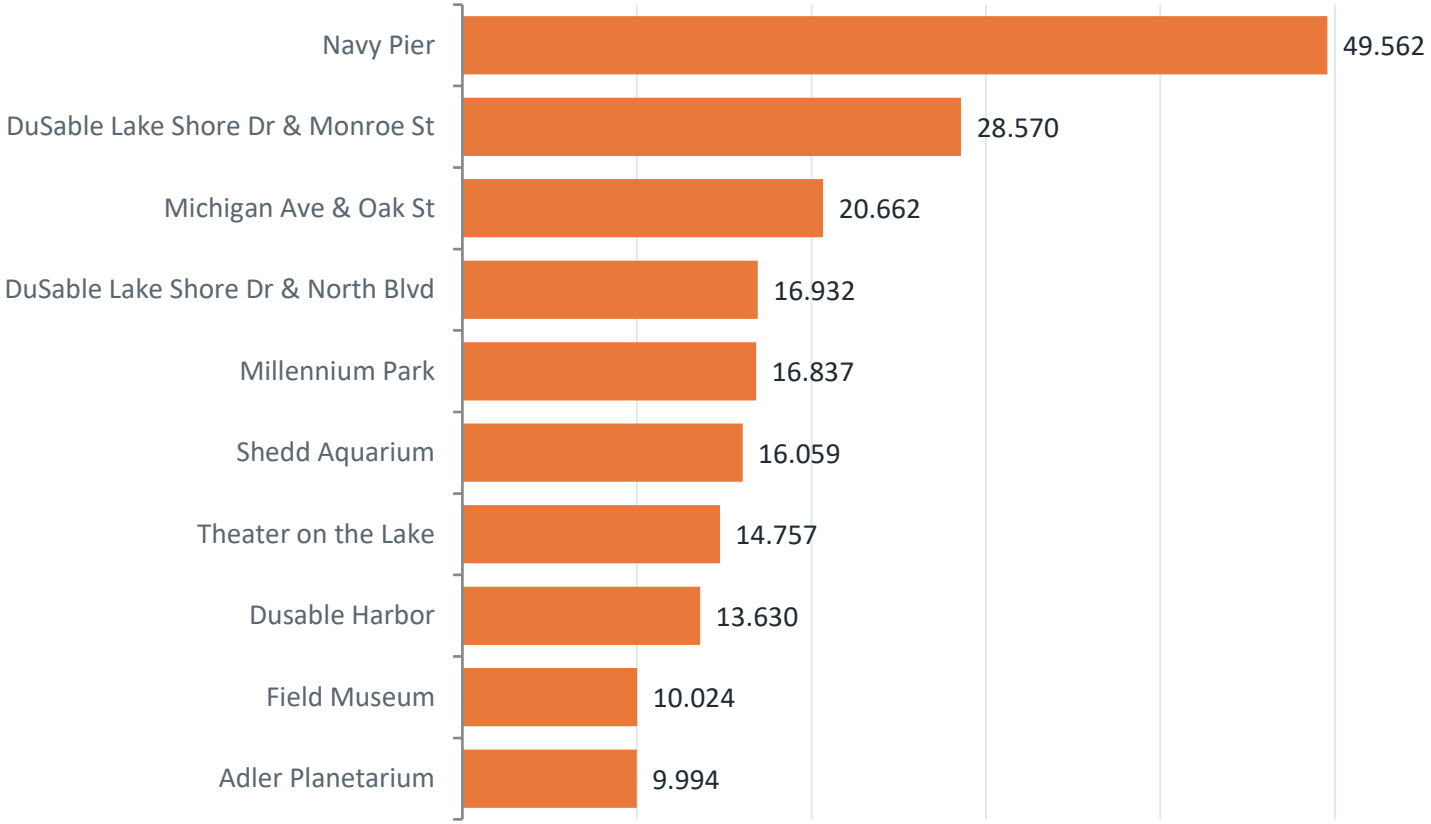
12 / 12 months

casual electric share is higher than members’ every month (3.7 to 10.1 points)

Rising

in both groups through the year; reason unknown

Casual trips concentrate at a few stations



Top 10 start stations for casual trips. Station data covers 80% of trips; the missing ones are electric bikes. Location interpretations are based on station names and were not verified.

0

stations shared with members in the top 10

11.7% · 6.1%

share of trips at the top 10 stations: casual and members

7.4% · 2.2%

round-trip share: casual and members (3.4×)

Limits of these findings

Trips, not users

No user IDs. Shares are shares of trips; the same person may have made many trips.

One year, no weather data

We can say “higher in warm months”, not “because of temperature”. Year-specific effects cannot be separated.

Station coverage is 80%

Missing stations are electric bikes. Results may skew toward classic bikes.

Duration filter was uneven

About 3.6% of casual trips and 2.1% of member trips were removed.

Purpose is not in the data

Timing and stations do not show why people ride or how likely they are to convert.

Factors not separated

Bike type and season were not controlled together. Some findings are exploratory.

Three recommendations: test with a small pilot first

1

Test a weekend or seasonal membership

Evidence: 52.9% of casual trips are Friday–Sunday. Their share is 17.9% in January and 40.8% in July.

Pilot: Offer the plan for a set period and compare conversion.

2

Pilot a campaign at busy casual stations

Evidence: The top 10 stations hold 11.7% of casual trips. Round-trip share is 3.4× the members'.

Pilot: Promote membership at a few stations and compare with control stations.

3

Start campaigns before summer

Evidence: Casual share is 27–29% in March–April and 37% in May.

Pilot: Start in April and measure the May–August effect.

The data shows where testing makes sense; only a pilot shows whether a campaign works.

Appendix: data source and method



Data and tools

Divvy public trip data (Motivate International Inc. license), 12 monthly tables. Analysis done in MySQL.



Method

Hypotheses, thresholds and refutation conditions were written before seeing results; later ones are marked “exploratory”. Every step was recorded in a cleaning log.



Note on tools

Queries were developed together with an AI assistant; results were verified with row totals and cross-checks.